INCH-POUND

MIL-C-70544A (AR)
AMENDMENT 5
17 February 1994
SUPERSEDING
AMENDMENT 4
24 February 1993

MILITARY SPECIFICATION

CARTRIDGE, 81MM, ILLUMINATING, M853A1,
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of MIL-P-70544A (AR), dated 28 January 1988, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

* 2.2, Change this paragraph to "2.1.2"

Under Product and Packaging Drawings, add the following:

"12944509 - Packing and Marking for Container, Ammo, PA156 and PA157; for Cartridge, 81mm"

Under Inspection Equipment Drawings, add the following:

"6518071 - Alignment Gage
6518072 - Alignment Check Gage"

PAGE 3

3.4, Change "60" to "65" and change "550,000" to "525,000"

3.4.1, Delete in its entirety and substitute the following:

"3.4.1 Pull test for oval sleeve crimp to suspension wire. The oval sleeve crimp to the suspension wire shall withstand a pull force of 1800 pounds minimum."

PAGE 7

4.4.1.d, Delete in its entirety and substitute the following:

"4.4.1.d Propelling charges, of one lot interfix number, from one manufacturer."

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PAGE 11

4.4.2.4, Under Major 105, delete in its entirety and substitute the following:

"Taped shroud loops not attached to suspension wire loop ...0.40% ...3.2 ...Visual."

Under Major category, add the new following:

"106 Tape missing from the sleeve and cable ...0.40% ...3.2 ...Visual."

PAGE 12

4.4.2.5, Change Major 107 from "Obturator ring not flush or below, Note 1 ...100% ...3.2 ...Visual" to "Entire obturator including tack weld not flush or below bourrelet diameter, Note 1 and 3 ...100% ...3.2 ...Gage"

Add new Major 112 as follows:

"112 Tail cone improperly assembled to the body tube assembly ...Note 4 ...3.2 ...4.4.5.11/Visual"

Add Note 3 and 4 as follows:

"Note 3. The inspection shall be performed at two places for each obturator, one at the tackweld and the other at 180 degrees from the tackweld.

Note 4. If a total of two or more defective units are found, the lot shall be rejected."

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* 4.4.2.8, Next higher assembly block: Add "/12944509" after "9362798"

Under Major category, add the following:

"104 Increment tape missing ...0.40% ...3.2 ...Visual
105 Propelling charge support missing or damaged ...0.40% ...3.2 ...Visual
106 Tab at the end of increment tape inadequate ...0.40% ...3.2 ...Visual
107 Tab at the end of increment tape covered by propellant charge support label ...0.40% ...3.2 ...Visual."

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* 4.4.2.9, Next higher assembly block: Add "/12944509" after "9362798"

PAGE 18

4.4.2.10, Change existing "Minor 202" to Minor 204".

Add new Minor 202, 203, 205 and 206 as follows:

"202 Plastic sheet not present in every sixth box ...0.65% ...3.2 ...Visual
203 Box containing plastic sheet not approximately marked ...0.65% ...3.2 ...Visual
205 Circular firing scale not present in every third box ...0.65% ...3.2 ...Visual
206 Box containing circular firing scale not approximately marked ...0.65% ...3.2 ...Visual"

PAGE 19a

Add new paragraph 4.4.2.12 (Classification of Defects & Tests) as follows:

* "4.4.2.12, Packing and Marking for Container, Ammo, PA156 & PA157 for Cartridge, 81mm (Prior to closing) sheet 1 of 1, Drawing Number 12944509:

Critical ...None defined

Major 101 ...Containers loose or improperly packed in metal can ...0.40% ...3.2 ...Visual/Manual

Major 102 ...Support assembly damaged, missing, or improperly placed ...0.40% ...3.2 ...Visual/Manual

Minor 201 ...Circular firing scale not present in every third metal box ...0.65% ...3.2 ...Visual

Minor 202 ...Ammo box containing circular firing scale not appropriately marked ...0.65% ...3.2 ...Visual

Minor 203 ...Plastic sheet not present in every sixth box ...0.65% ...3.2 ...Visual

Minor 204 ...Evidence of poor workmanship ...0.65% ...3.7 ...Visual"

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PAGE 19b

* Add new paragraph 4.4.2.13 (Classification of Defects & Tests) as follows:

"4.4.2.13, Packing and Marking for Container, Ammo, PA156 & PA157 for Cartridge, 81mm (After closing):

Critical ...None defined

Major 101 ...Contents of box loose ...0.40% ...3.2 ...Manual

Major 102 ...Cover assembly not secure ...0.40% ...3.2 ...Manual

Minor 201 ...Marking missing, misleading or unidentifiable ...0.65f% ...3.2 ...Visual

Minor 202 ...Evidence of poor workmanship ...0.65% ...3.7 ...Visual"

PAGE 20

4.4.3.2, Change "60" to "65" and change "550,000" to "525,000"

PAGE 22

4.4.3.3, Change "Major: Illuminant fails to ignite or provide effective illumination for less than 50 seconds (see Note 2)." to "Major: Illuminant fails to ignite or provide effective illumination for less than 50 seconds in the air (see Note 2 & 3)".

Note 2, Delete in its entirety and substitute the following:

"If the height above ground level at which the tail cone separated from the tube body was less than 300 meters, the performance of the cartridge will not be considered in evaluating the acceptability of the lot for illumination time defects."

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Add Note 3 as follows:

"Note 3: The following are considered some causes of this defect:

a. Streaming chute/ripped chute.
b. Partial or fully tangled chute.
c. Chute fails to deploy.
d. Illuminant detached from canister (illuminant dropout).
e. Parachute detached from canister."

PAGE 23

4.4.3.5, Change "8865018" to "9209555-8".

Delete "4.4.3.5.1 and 4.4.3.5.2" and substitute "4.4.5.10".

4.4.3.5.1, Change "1050" to "1800".

At the last sentence, delete "until it is requalified in accordance with paragraph 4.4.3.5.2."

4.4.3.5.2, Delete in its entirety.

PAGE 26

4.4.5.6, Delete the word "primers" from the fifth line. Add "and chamber pressure" after muzzle velocity in the last sentence before Table III.

PAGE 27

4.4.5.11, Add new paragraph as follows:

"4.4.5.11 Tail cone and body tube inspection. At the end of each assembly lot produced, fifty random samples shall be selected for this inspection. The projectile loading assemblies shall be subjected to a pull load sufficient to shear the pins; slowly remove the body tube assembly from the tail cone and once the canister is visible, hold the canister down to prevent movement from the tail cone while removing the body tube. Inspect the pull cord for the following: Pull cord should be coiled about one half of the length and inserted into a 3/4" loop attaching the pull cord to the parachute bag and the other one half should be between the parachute disc and the droque plate. If the pull cord does not meet the above or is tangled or pinched, the sample shall be classed as defective. In addition, if the cable is frayed, kinked or deformed, the sample shall be classed as defective."

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* 5.2.1, Add "/12944509" after "9362798".

* 5.3.1, Add "/12944509" after "9362798-1".

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